

INFERRING CARBON DIOXIDE EMISSIONS FROM POWER PLANTS USING SATELLITE IMAGERY AND MACHINE LEARNING

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ABSTRACT

Emissions from fossil fuel power plants are a major contributor to climate change. Directly measuring emissions at the source is cost-prohibitive throughout most of the world, while self-reporting varies widely in detail, recency, and spatiotemporal resolution. We use machine learning to infer power generation and carbon dioxide (CO₂) emissions from proxy signals in multi-spectral satellite imagery, including Landsat 8, Sentinel-2, and PlanetScope. We built and evaluated classification models to predict plant on/off status and regression models to predict generation. By training on a data set of power plants for which we know the generation, we are able to apply our models globally with higher spatial and temporal precision than alternative approaches, producing a scalable and consistent approach to estimate CO₂ emissions.

Index Terms— machine learning, remote sensing, fossil fuel power plants, greenhouse gas emissions, climate change

1. INTRODUCTION

Responding to climate change requires timely and accurate measurement of greenhouse gas (GHG) emissions, especially CO₂. The Paris Agreement, adopted in 2015, set goals to limit global temperature rise and established frameworks for nations to report on GHG emissions and steps to reduce them [1]. The energy sector is responsible for the majority of GHG emissions – 76% globally as of 2019, representing 38 GtCO₂e [2]. Nearly half of the energy sector's GHG emissions result from electricity generation, accounting for 16 GtCO₂e in 2019 or over 31% of total global GHG emissions [2].

Power plant emissions monitoring and reporting currently includes continuous emissions monitoring systems (CEMS) and bottom-up approaches [3]. CEMS measures emissions at individual plants, providing reliable measurements. However, CEMS are costly with limited deployment globally and require continual calibration [4]. A more common approach is bottom-up self-reporting, which quantifies emissions at

power plants using fuel consumption, fuel quality, and emission factors [5, 6]. Yet the detail, recency, and spatiotemporal resolution of self-reporting varies by region, posing challenges for policymakers to design sustainability strategies [7, 4]. In addition, GHG monitoring satellites and aerial surveys have been used to monitor power plant emissions [7, 4]. However, existing approaches lack spatiotemporal resolution or have limited deployment. Alternatively, multi-spectral satellites that observe Earth's surface provide an opportunity to improve monitoring. Combining imagery from these satellites provides persistent global coverage and can identify more granular power plant activity.

To complement current emissions monitoring approaches, we developed a method that uses multi-spectral satellite imagery paired with ground truth generation data to train machine learning (ML) models to infer a power plant's on/off status and generation. Our project is part of the initiative called Climate TRACE (Tracking Real-time Atmospheric Carbon Emissions; <http://climatetrace.org>), which provides GHG emissions estimates for all sectors globally. We trained models on plants in countries with hourly or sub-hourly generation data by assimilating data from multiple sources, then applied these models globally using region-, plant-type-, and fuel-specific carbon intensities to convert plant-level generation estimates to CO₂ emissions.

2. MATERIALS AND METHODS

We trained ML models on multi-spectral satellite imagery to infer a power plant's operational status based on the visible emitted water vapor plumes from cooling towers and flue stacks (Fig. 1), since the amount of heat released is directly related to power generation at the time. We estimated generation as opposed to CO₂ directly because the vapor plume is more strongly associated with generation than with emissions, as fuel characteristics and power plant equipment varies, which impacts the total amount of GHG emissions.

Two common power plant structures which emit vapor

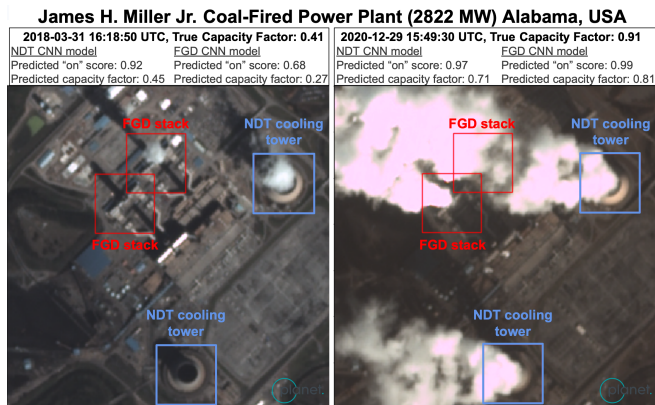


Fig. 1. PlanetScope CNN predictions on the James H. Miller Jr. power plant at low vs. high generation. Separate NDT and FGD models predict on NDT cooling tower (blue) and FGD stack (red) patches, respectively. NDT plumes are generally wider than FGD due to the different size and shape of the vapor outlets. These predictions are ingested by subsequent models to estimate generation, then CO₂, for the plant [8].

plumes visible in multi-spectral satellite imagery are natural draft wet cooling towers (NDT) and wet flue gas desulfurization (FGD) stacks. Plants using NDT produce visible vapor plumes during cooling. FGD removes sulfur dioxide from gases released at coal-fired power plants, wherein flue gases become saturated with water, producing visible plumes. A power plant may have one, both, or neither of these technologies. We created separate models for NDT and FGD due to differences in size and shape of resulting plumes, see Fig. 1.

2.1. Data Sources

We sourced satellite imagery for model development from: 1) Landsat 8¹ Collection 2 Tier 1 (top-of-atmosphere) at a 16-day revisit rate, aerosol to shortwave infrared (SWIR), and thermal IR bands at 30m and 100m spatial resolutions, respectively (0.43 to 12.51 μm); 2) Harmonized Sentinel-2² Level 1C (top-of-atmosphere) at a 5-day revisit rate, aerosol, visible to near infrared (NIR), and SWIR bands at 60m, 10m, and 20m spatial resolutions, respectively (0.44 to 2.19 μm); 3) Planet Labs PlanetScope³ imagery at a daily revisit rate, visible to NIR (0.46 to 0.89 μm) and 8 additional Usable Data Mask (UDM2) image quality bands at 3m spatial resolution.

When possible, our models used all satellite bands and derived products available, plus some additional bands we created through post-processing: haze-optimized transformation (HOT; a linear combination of blue and red bands), whiteness (summing RGB bands), normalized difference vegetation index (NDVI), and normalized differences between the

two SWIR or thermal bands where applicable. These additional derived bands provided beneficial features for some of our models, where the HOT and whiteness acted as a plume mask. Our gradient boosted tree models used all these bands, while the neural networks were limited to RGB to enable transfer learning, as described in Section 2.2.

We created a harmonized power plant inventory containing information on location, fuel type, capacity, operating dates, cooling type, and pollution control technology at both the plant and unit level by assimilating data from a variety of sources, detailed further in our 2022 methodology document [8]. We used OpenStreetMap to label flue stacks and cooling towers at all power plants in our data set, enabling us to create patches around the structures most relevant to the modeling task. In order to train models to predict plant utilization at the time satellite imagery was captured, we required at least hourly generation data to match up to the imagery for training, which we sourced from the USA⁴, Europe⁵, and Australia⁶.

We further restricted the training data to optimize for both modelability and impact: 1) plants with capacity >500 megawatts (MW) of which coal accounted for $\geq 50\%$ and gas <50%; 2) for NDT models, $\geq 70\%$ of the plant's cooling was NDT; 3) for FGD models, $\geq 90\%$ of total generation and 100% of coal capacity was associated with wet FGD; 4) the cloud mask over the plant indicated <20% cloud coverage; 5) to limit to conditions in which we expect water vapor to be visible for FGD models, $T < 25^\circ\text{C}$ and $H - T > 15\%$ for a temperature in $^\circ\text{C}$ (T) and humidity percentage (H) with weather data obtained from World Weather Online⁷. For PlanetScope, we used the post-2018 UDM2 bands to keep only images with <80% heavy haze, as well as calculated mean brightness and developed a cloudiness score based on HOT, whiteness, and NDVI to respectively filter out excessively dark or cloudy images. Finally, we excluded from training any plants or images with observed data quality issues. Collectively, these filters resulted in 55 training plants for NDT and 76 for FGD for the years 2015-2022, with the number of training images varying by sensor: 38 765 for PlanetScope NDT, 38 395 for PlanetScope FGD, 10 813 for Sentinel-2 NDT, 10 159 for Sentinel-2 FGD, 5 312 for Landsat 8 NDT, and 4 415 for Landsat 8 FGD.

2.2. Modeling Approaches

To estimate power plant generation and CO₂ emissions, we built models to identify power plant activity. These models performed the following tasks given a satellite image at a certain point in time: 1) classification to predict if a plant

⁴US EPA Clean Air Markets Program Data, <http://campd.epa.gov>

⁵European Network of Transmission System Operators for Electricity, <http://transparency.entsoe.eu/generation/r2/actualGenerationPerGenerationUnit/show>

⁶National Electricity Market, <http://aemo.com.au/energy-systems/electricity/national-electricity-market-nem/about-the-national-electricity-market-nem>

⁷<http://worldweatheronline.com/developer/api>

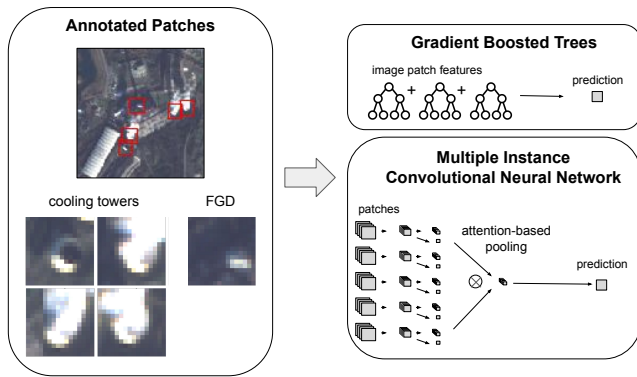


Fig. 2. Overview of modeling approaches: NDT cooling towers and FGD stacks were cropped from satellite imagery and fed into ML models to classify the power plant’s on/off status and predict the capacity factor.

was generating or not (on/off) and 2) regression to estimate the capacity factor (generation divided by capacity). We included both sets of models because the on/off task is simpler and can be predicted more accurately, while the regression task is essential for differentiating low from high generation. We multiplied predicted capacity factor by capacity to infer generation. We trained two different types of models: gradient-boosted decision trees and convolutional neural networks (CNNs), building separate models for NDT and FGD, as well as for each satellite data set.

To focus our models on the most relevant parts of the plant, we used the annotated NDT cooling tower and FGD stack patches as the inputs (Fig. 2). This strategy was previously shown to produce more accurate classification models than a single image of the entire plant [9]. The patch size around each tower or stack is optimized as a hyperparameter for each model type and imagery product.

The gradient-boosted decision tree models used XGBoost. Features were calculated by taking the mean, standard deviation, and 90th percentile of the pixel values in each imagery band of patches of varying sizes centered on FGD or NDT structures, producing a vector of statistics which we aggregated to the image level using mean, min, and max operations.

The CNN models used a multiple instance framework to combine the patches through an attention layer to aggregate features from the different patches for a plant [9]. We encoded each patch with a CNN (truncated after some of the convolutional blocks). We used transfer learning to initialize model weights with a ResNet50 CNN [10] pre-trained on the RESISC data set, which consists of aerial RGB images labeled by land cover and scene classes [11]. For classification, we added a softmax layer and used cross-entropy loss; for regression, we added a sigmoid layer and used either mean squared error or Huber loss based on performance. While the XGBoost models used all satellite bands, our CNN models

Table 1. On/off classification (Cls.) and capacity factor regression (Reg.) out-of-sample performance for 2015-2022.

Imagery	Model	NDT		FGD	
		Cls. mAP	Reg. RMSE	Cls. mAP	Reg. RMSE
PlanetScope	XGBoost	0.98	0.21	0.93	0.28
PlanetScope	CNN	0.97	0.21	0.91	0.25
Sentinel-2	XGBoost	0.98	0.21	0.93	0.25
Sentinel-2	CNN	0.98	0.19	0.93	0.25
Landsat 8	XGBoost	0.92	0.24	0.88	0.28
Landsat 8	CNN	0.89	0.29	0.83	0.28

only used the RGB bands in order to make use of transfer learning from RESISC. More details about these models are provided in [8].

Model performance was determined using four-fold cross validation. We built a regular $1^\circ \times 1^\circ$ grid in latitude-longitude space and placed plants in the same grid cell into the same fold to ensure no plant occurred in multiple folds to prevent data leakage. For each iteration, two folds of plants were used for training, one for validation, and one for testing.

We independently tuned the hyperparameters of both XGBoost and CNN models to optimize for validation mean average precision (mAP) and root mean squared error (RMSE) for classification and regression, respectively. We selected mAP due to the class imbalance; plants were “on” in about 80% of the training images.

Next, a generation ensemble model aggregated the predictions from both models for all three sensors into an integrated estimate of plant utilization. Lastly, we applied carbon intensities specific to the region, fuel, and power plant type to convert generation to CO₂ emissions. More details and results regarding emissions modeling are provided in [8].

3. RESULTS

Model performance for 2015-2022, displayed in Table 1, was calculated on test predictions aggregated across all four cross-validation folds. The naïve baseline mAP is 0.5 for a model that predicts the most prevalent class, or always “on.” The regression model’s naïve baseline predicts the training set’s mean capacity factor, resulting in an RMSE of 0.33 for NDT and 0.34 for FGD. All of our models outperformed these baselines, especially on the simpler task of classification. NDT models generally did better than FGD because NDT tends to produce larger plumes. PlanetScope and Sentinel-2 models performed comparably, while the Landsat 8 models generally performed worse perhaps due to the coarser (30m) spatial resolution. However, the results show that Sentinel-2’s 10m resolution is sufficient and PlanetScope’s 3m resolution did not substantially improve model performance. Still, PlanetScope’s daily revisits offer a major advantage in temporal resolution over Sentinel-2 and Landsat 8, resulting in a higher number of total observations over weeks, months, and years.

Table 2. 2021 CO₂ estimates using the ML approach for the five highest-emitting power plants in the world with NDT or FGD. CO₂ emissions include fossil and biogenic fuels.

Power Plant	Country	Capacity (MW)	2021 Estimated Capacity Factor	2021 Estimated Emissions (MtCO ₂)
Tuoketuo	China	6720	0.60	31.6
Beilun	China	5160	0.58	23.6
Drax	UK	3997	0.55	22.1
Ninghai	China	4400	0.64	22.1
Majuba	South Africa	4143	0.60	22.0

4. DISCUSSION

Monitoring individual power plants by applying ML models to satellite imagery makes it possible to build a picture of generation activity and emissions that is more uniform, recent, and detailed than other reported data. Our approach can be applied globally, including to regions where little to no generation or emissions data are publicly reported. By expanding publicly-available power plant emissions estimates on a frequent and plant-wise basis, this work can inform governments, corporations, and citizens to develop sustainability strategies for greater impact.

Our CO₂ emissions estimates for power plants across the world are hosted by Climate TRACE (<http://climatetrace.org>). The inventory currently contains country-level annual emissions estimates for 2015-2021 and plant-level annual CO₂ estimates for 2019-2021 for the top 500+ highest-emitting power plants. Of these, ~200 were estimated using the approach described in this paper, representing <1% of power plants but responsible for roughly 17% of global power plant CO₂ emissions in 2021. Table 2 shows the five highest-emitting plants estimated using this approach for 2021.

We continue to refine and improve the accuracy and coverage of our predictions in an effort to provide plant-level emissions estimates for more power plants. This includes improving our regression models by better understanding the relationship between plume size, generation, and weather conditions; creating mechanisms to estimate model uncertainty and bias; including new satellite observations; sourcing additional ground truth data from regions outside the current training set to both validate and mitigate model bias; and investigating new proxy signals at plants that do not use NDT or FGD as well as signals widely-applicable to other fuel sources. This research is part of an ongoing project; the next Climate TRACE inventory update is scheduled for Fall 2023 to include more power plants and 2022 data.

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